

Agenda

- BI project beginning (CC)

Date	Subject
17.sept	Introduction to BI / Data Warehousing & Data Modelling
24.sept	Data Modelling (Practice)
01.oct	ETL
08.oct	Cube OLAP & Analysis
15.oct	Finland week / No course
22.oct	BI Project
29.oct	IA & BI / BI Project
05.nov	Data Analysis / BI project
12.nov	BI project
19.nov	BI project / Data Analytics (Schumacher)
26.nov	Data Analytics (Schumacher)
03.déc	Data Analytics (Schumacher)
10.déc	Data Analytics (Schumacher)
17.déc	Data Analytics (Schumacher)
24.déc	Data Analytics (Schumacher)

> The project setting



- Mantis (a widely used Bugtracker)
  - http://www.mantisbt.org/
- Scribus: Open Source: page layout program
  - http://www.scribus.net/canvas/Scribus



- Scribus uses Mantis
  - http://bugs.scribus.net/my_view_page.php

> Our data source

- Goto
  - http://teachingse.hevs.ch/csvFiles/
- There you find a snapshot of a status of the bug tracking system
- Note: Web Access only possible inside the HES-SO network
- Every Monday (6 am) a new file will be uploaded

> Group Work

Groupwork: A group contains of 2 people!

Remember: 60% of the CC (50% of the module)

You can use async time & lecture time

> How will the project be graded

- We will have weekly online(?) discussions
  - Presence is expected
  - We will discuss your current progress
  - And next challenges
  - You will be given a projection of your grade (mainly based on continuous project progress)

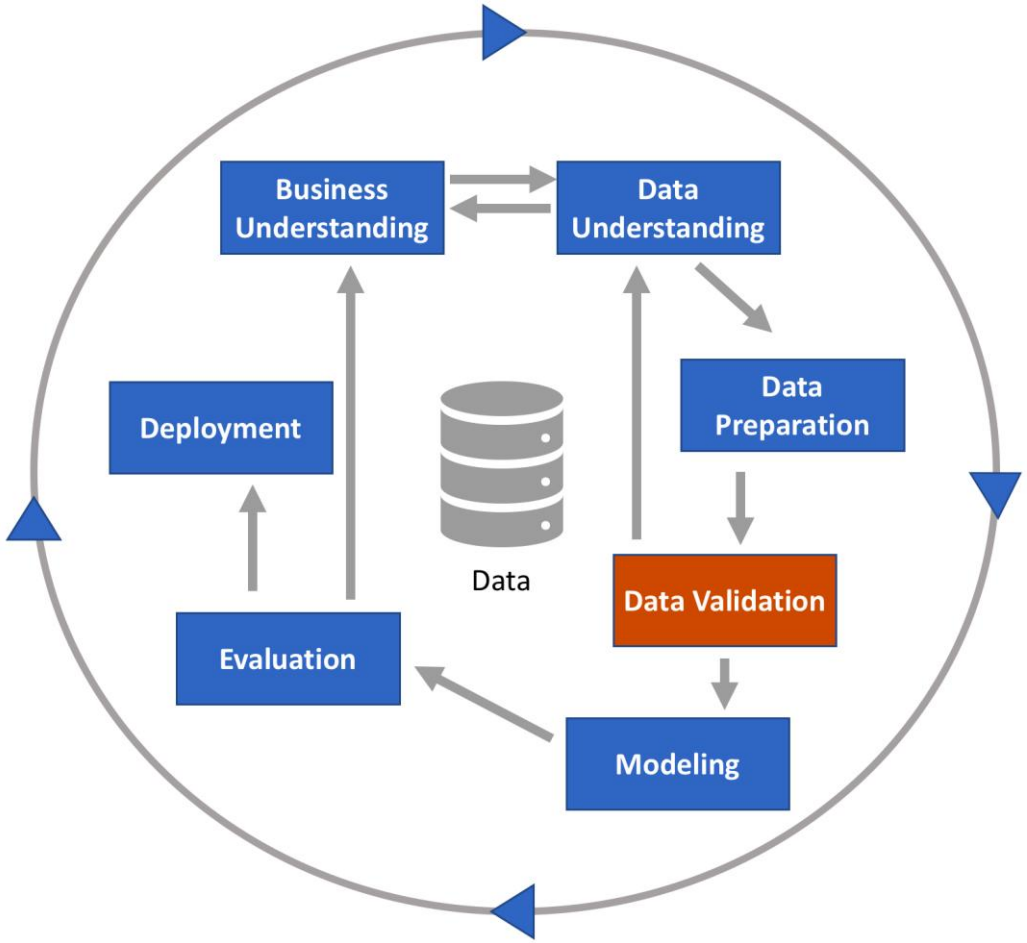
- Criteria for grading
  - Continuous project progress
  - Final product
  - Documentation

Project Timelines

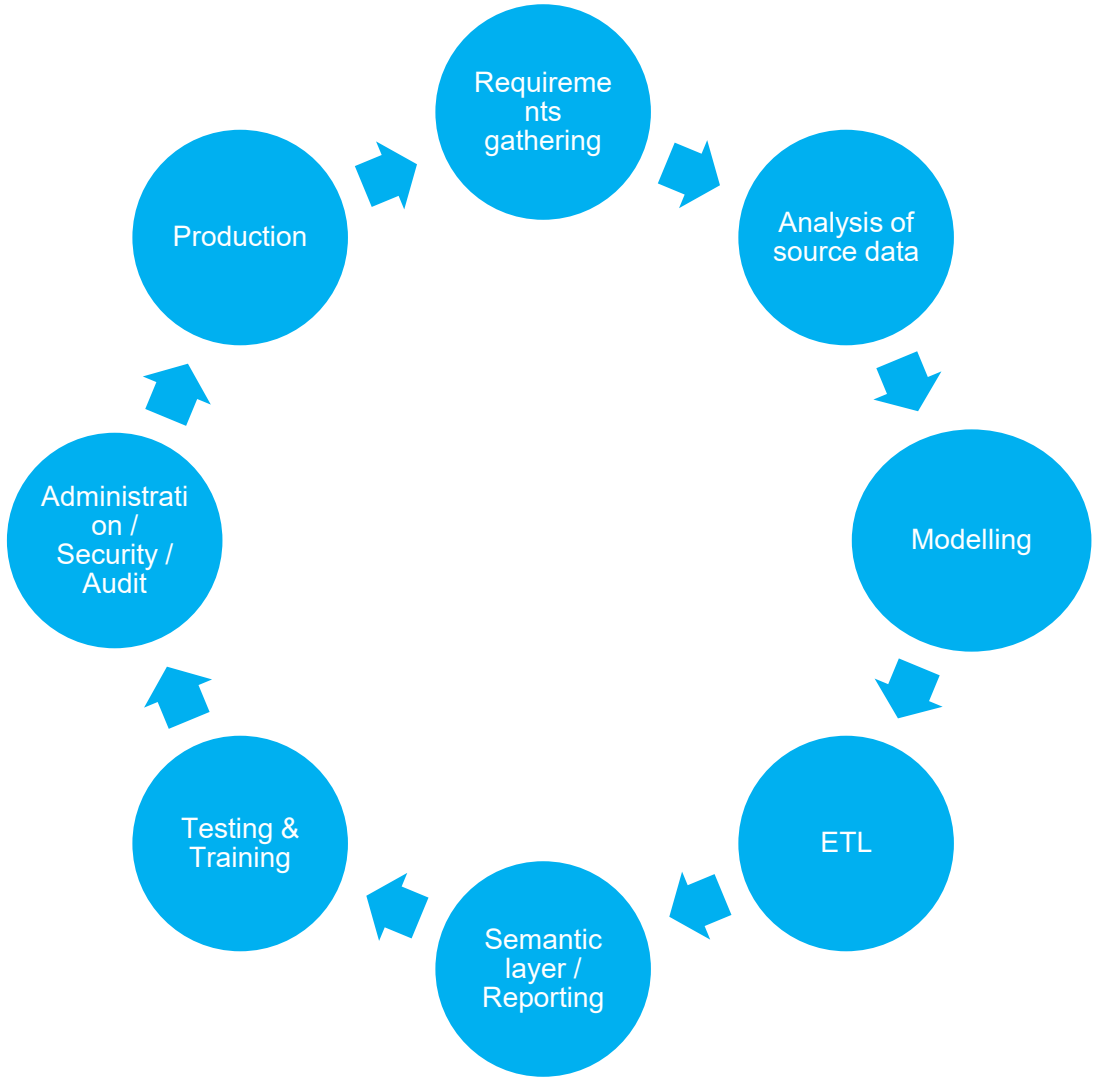
1. Make yourself an overview about the data (22.10)
2. Which questions do you want to answer? (22.10)
3. Create a database schema for the warehouse (Have in mind that we need to update the data! (29.10)
4. Create the data warehouse (29.10)
5. Apply the ETL process (5.11)
6. Provide means for updating the warehouse based on new snapshots (12.11)
7. Automate the update process (19.11)
8. Create a cube / reports answering your questions (19.11)
9. Provide a documentation, about your project (along with your project) (19.11) & THERE IS NO PRESENTATION

BI Management
Project stages

- Iterative development
- User integration
- Data testing and validation

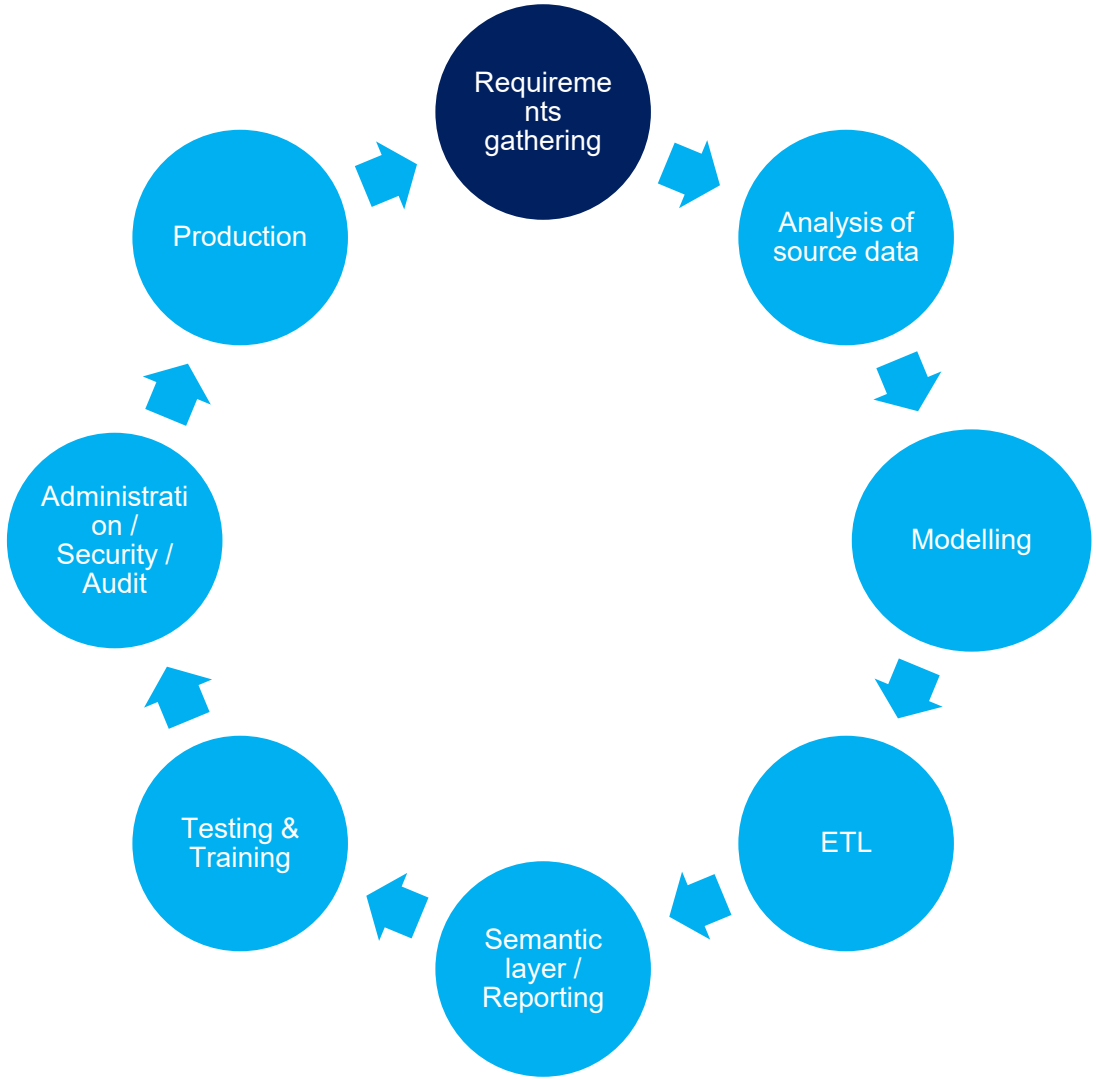


BI Management
Project stages



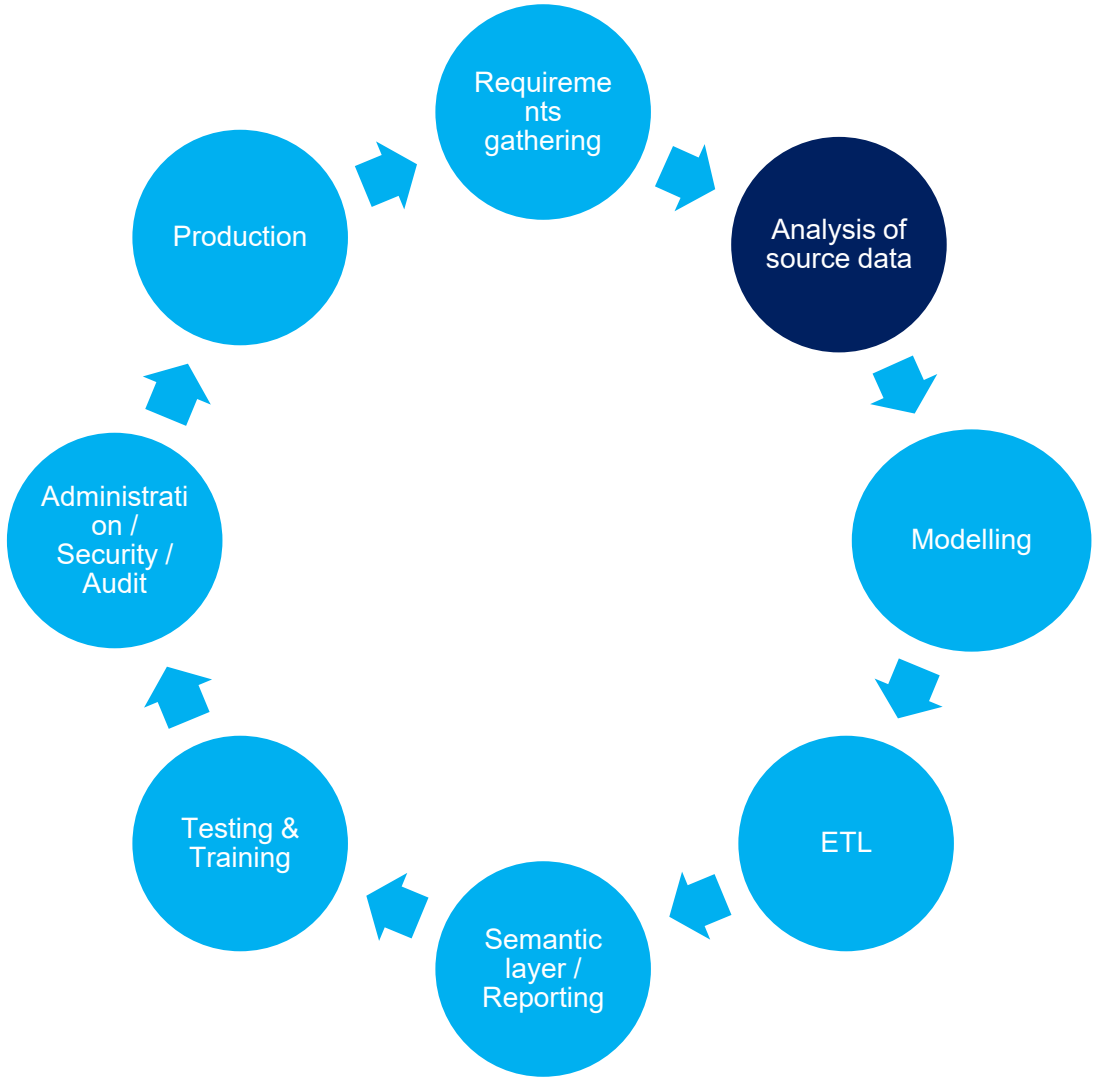
BI Management
Project stages

- Requirements gathering
  - Select key users (business and IT)
  - Identify requirements (interviews with the business)
  - Identify data (interviews with IT)



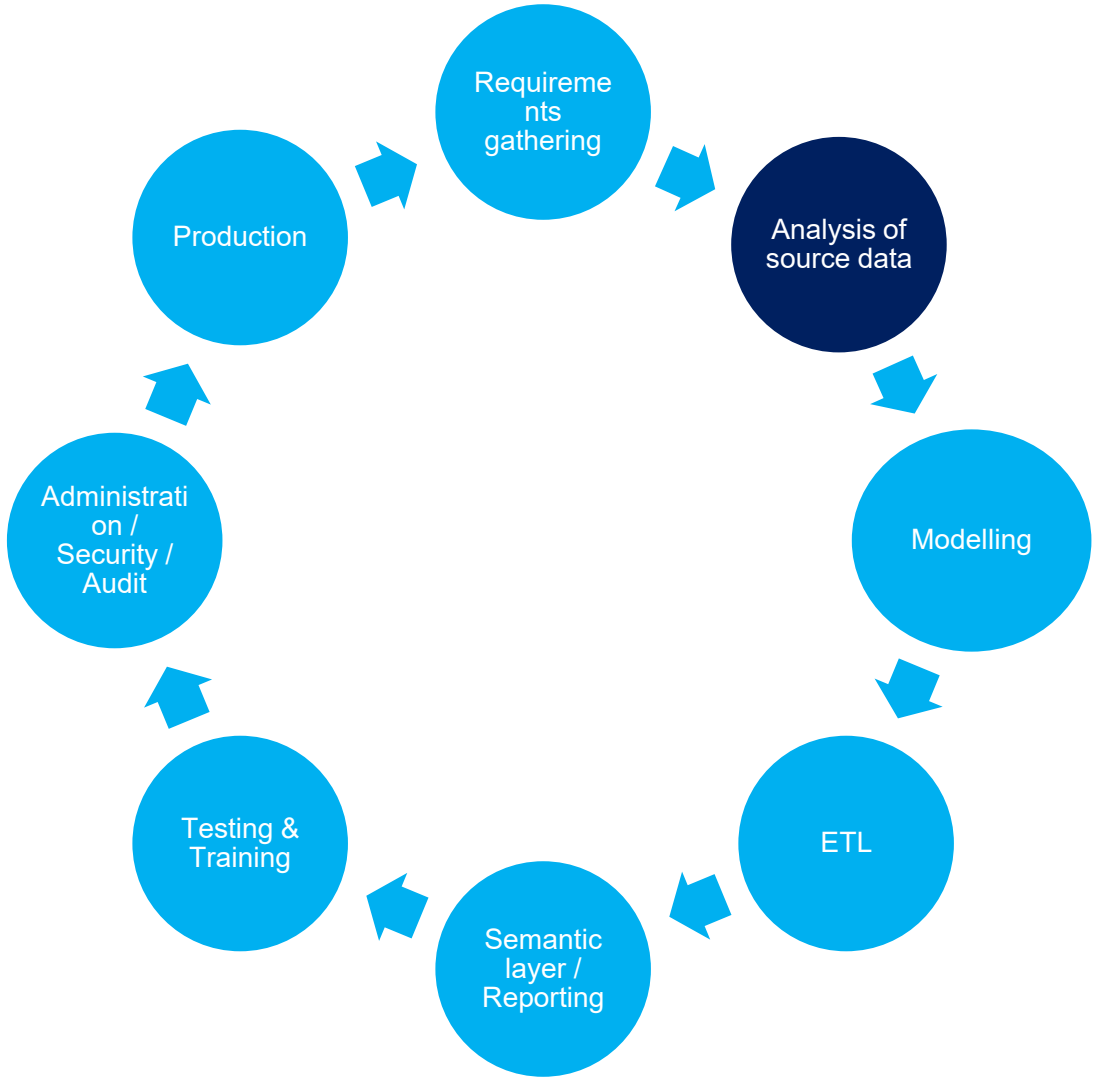
BI Management
Project stages

- Source data analysis
  - Data quality
  - Available history
  - Management rules
  - Fields to be archived



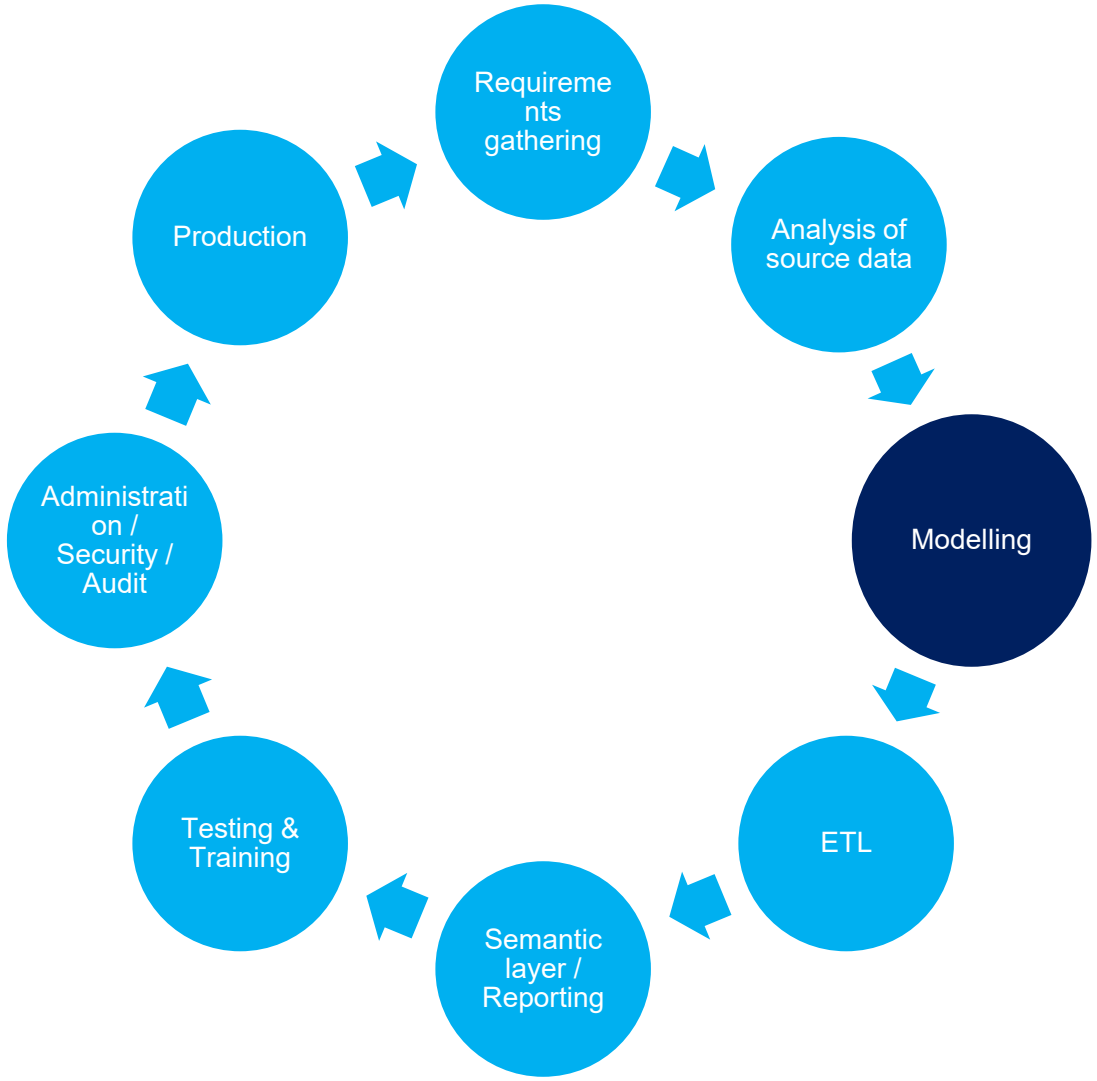
BI Management
Project stages

- Modelling
  - Define fact tables based on granularity
  - Create dimensions (denormalisation)



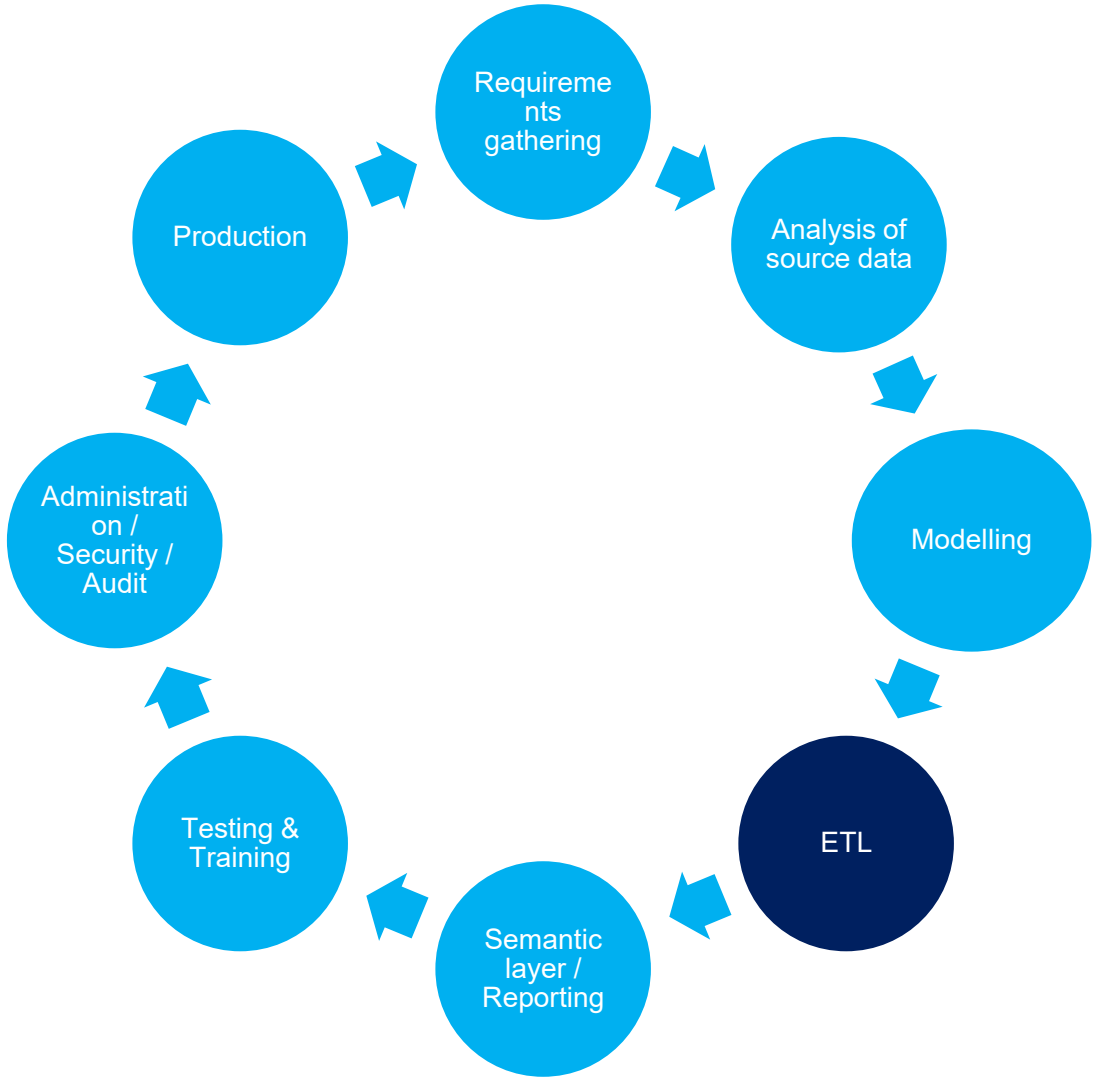
BI Management
Project stages

- Modelling
  - Define fact tables based on granularity
  - Create dimensions (denormalisation)



BI Management
Project stages

- ETL
  - Extract Transform Load
  - Data retrieval (Full/Delta)
  - Processing
  - Loading into target tables
  - Load frequency
  - Mapping document



BI Management
Project stages

- Semantic layer
  - SQL code generator that masks the technical aspects of the database
  - E.g.: Universe & DataSet

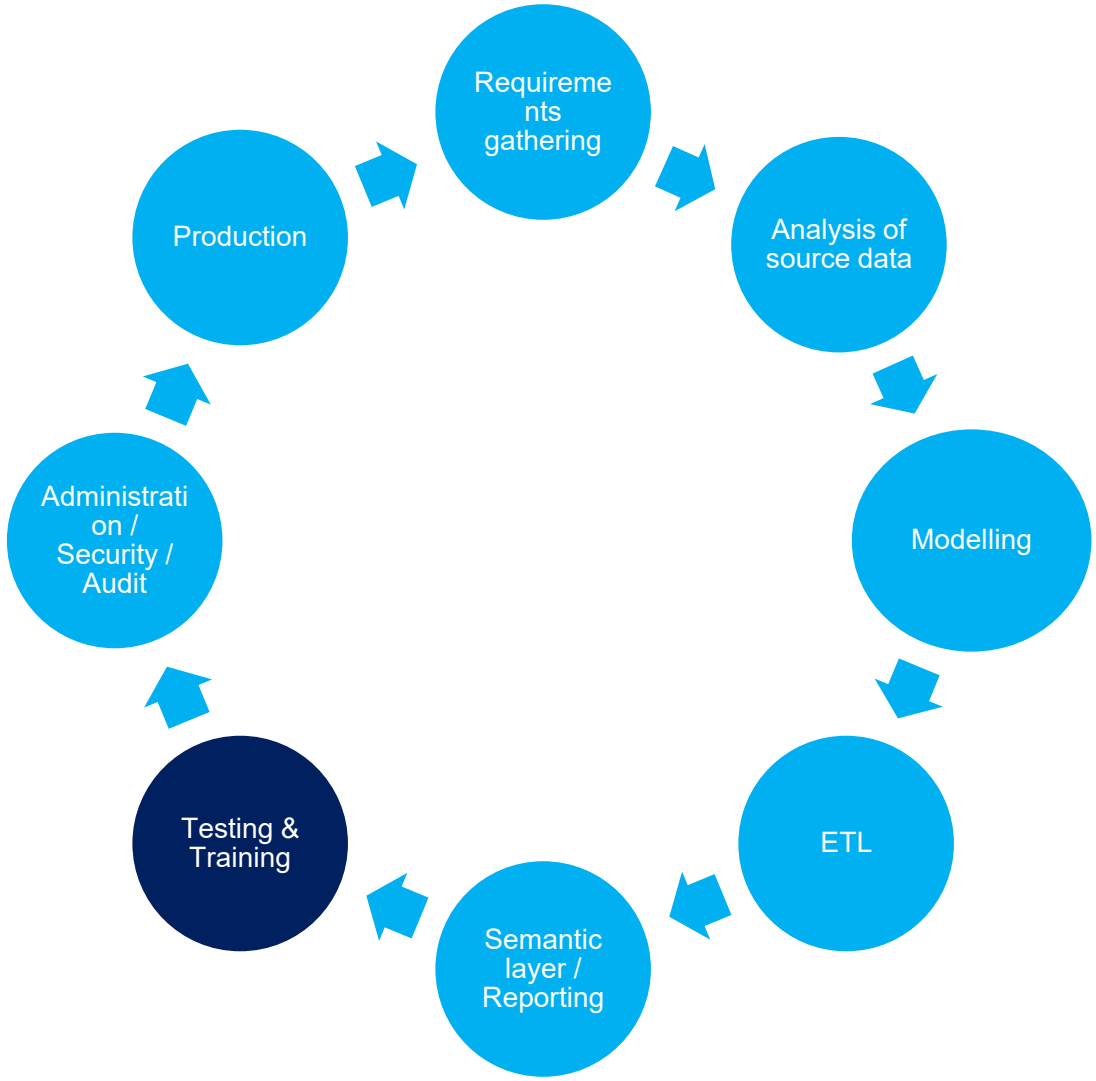
- Reporting
  - Report creation
  - Document template creation



BI Management
Project stages

- Tests
  - With business and IT key users

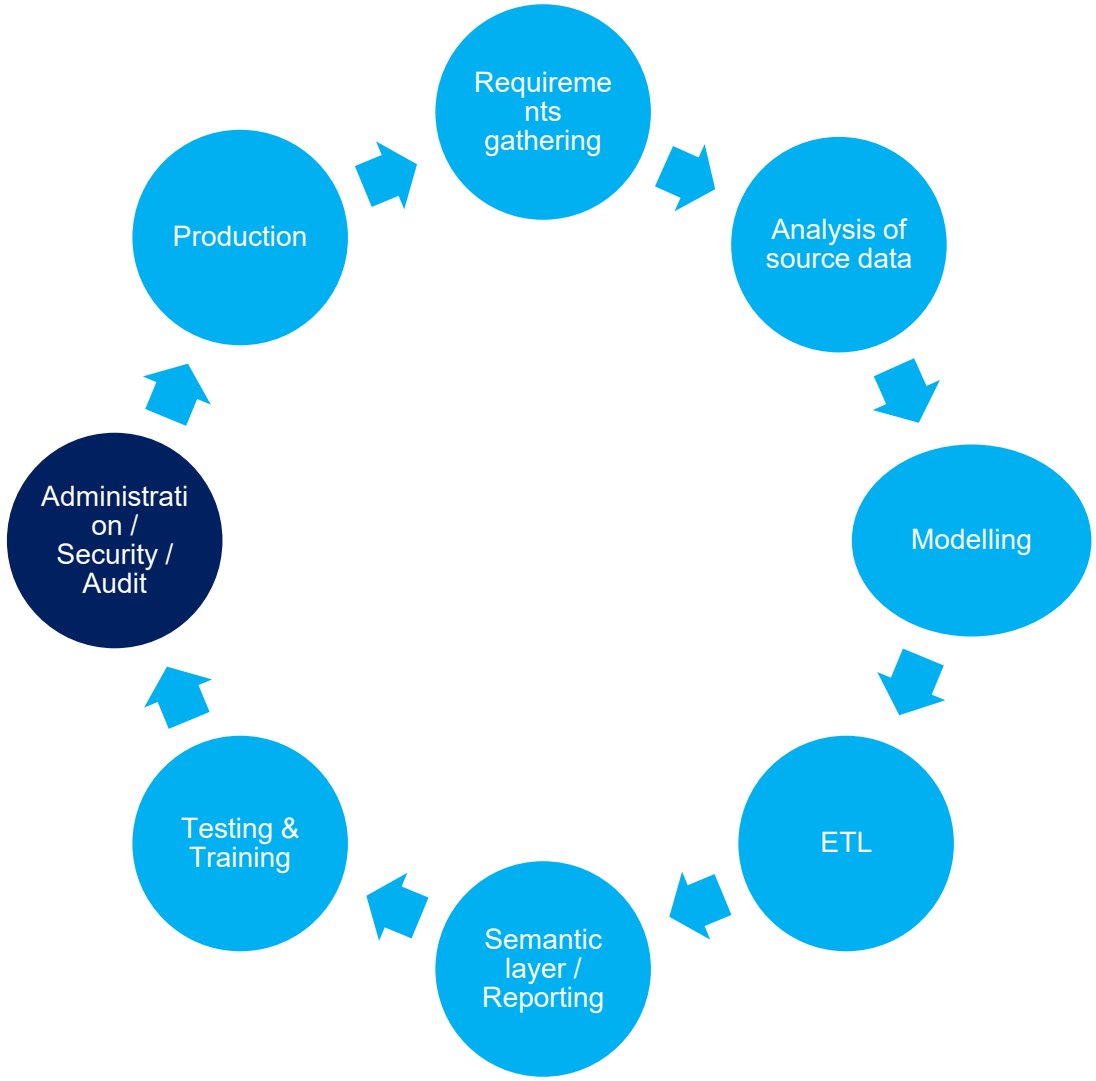
- Training
  - Reporting for end users
  - Administration for IT



BI Management
Project stages

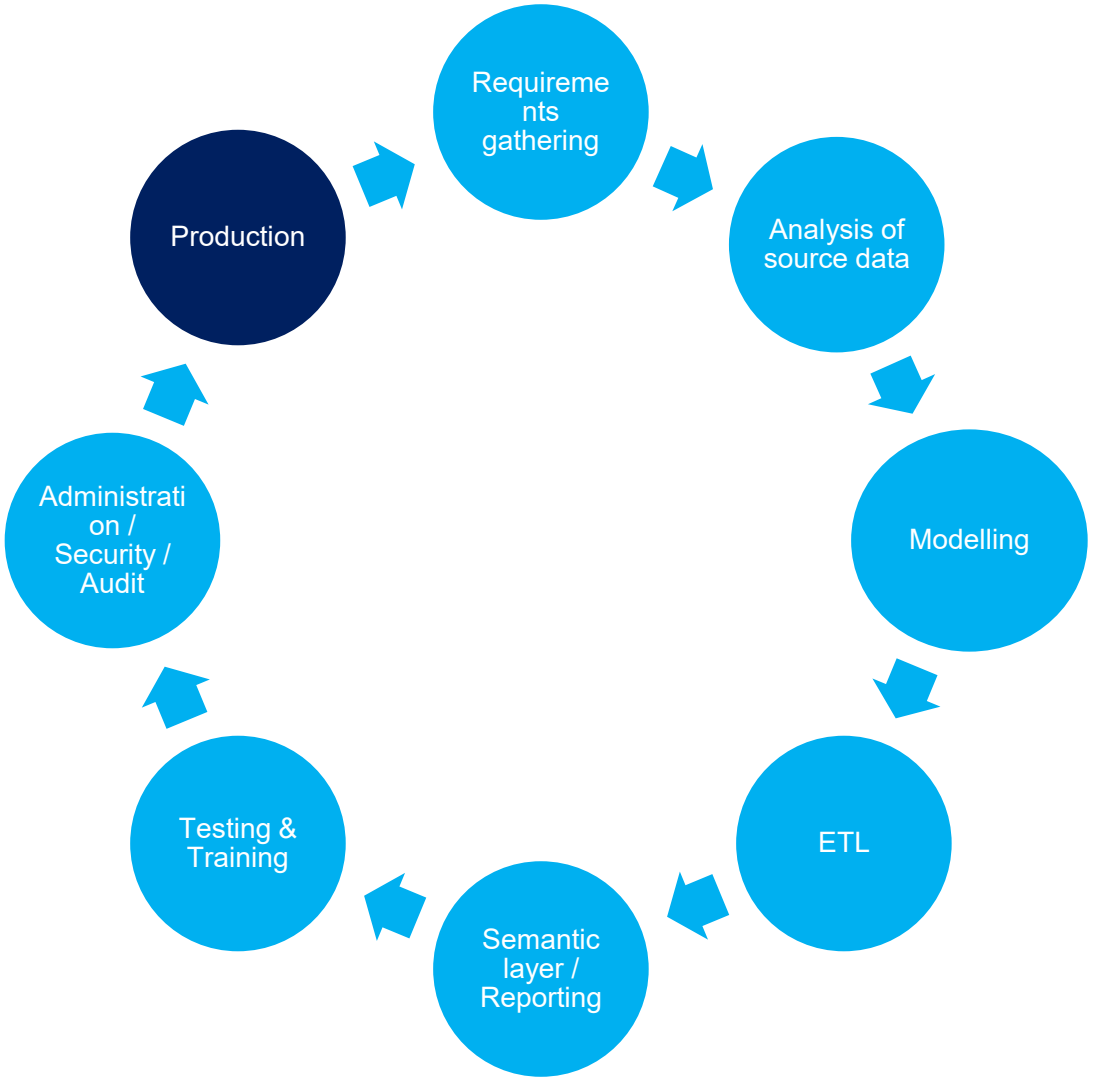
- Administration
  - User and group creation
  - Authentication mapping
  - Data security

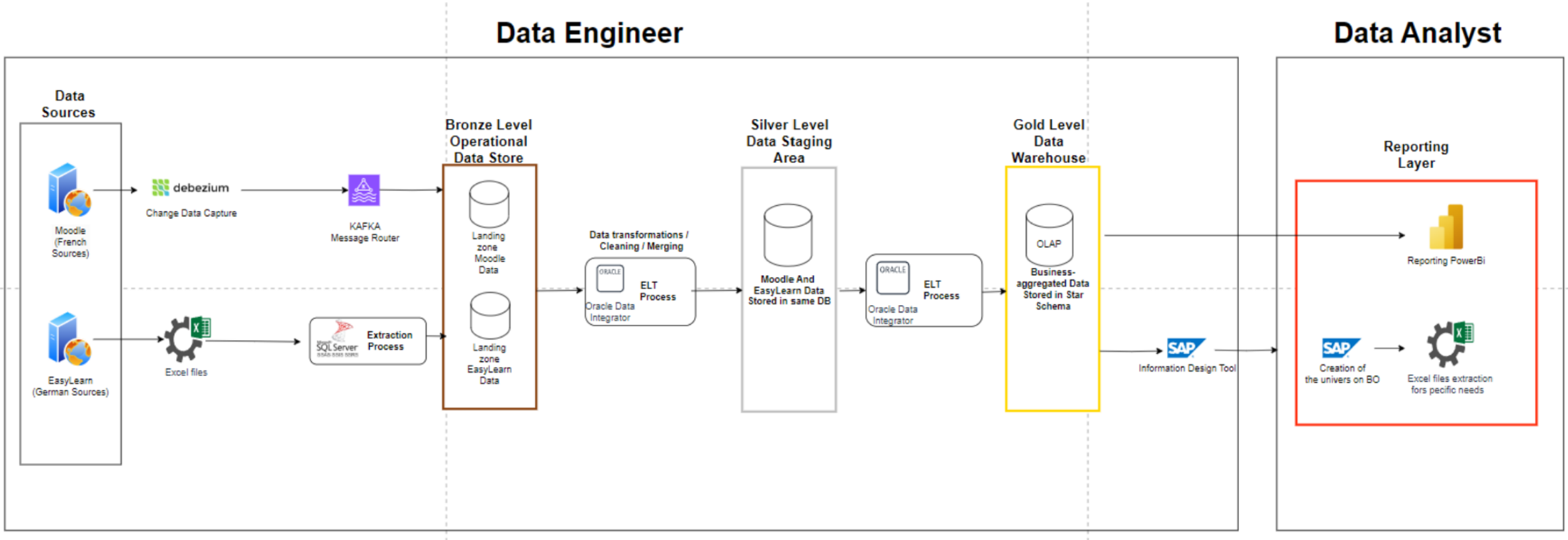
- Audit
  - BI server activity
  - BI server content (repository)



BI Management
Project stages

- Production launch
- Delivery
- Documentation
- Load automation
- Report automation




Data Engineer
Data Analyst
Data Sources
Moodle (French Sources)
EasyLearn (German Sources)
debezium
Change Data Capture
KAFKA Message Router
Excel files
Extraction Process
Bronze Level Operational Data Store
Landing zone Moodle Data
Landing zone EasyLearn Data
Data transformations / Cleaning / Merging
Oracle Data Integrator
ELT Process
Silver Level Data Staging Area
Moodle And EasyLearn Data Stored in same DB
Oracle Data Integrator
ELT Process
Gold Level Data Warehouse
OLAP
Business-aggregated Data Stored in Star Schema
Information Design Tool
Reporting Layer
Reporting PowerBi
Creation of the univers on BO
Excel files extraction fors pecific needs

BI elements
Introduction

Source systems

Extract

Extract

Extract

Data Preparation Area

- Storage (ODS)
  - Flat files
  - RDBMS
  - Other
- Processing (DSA)
  - Cleaning
  - Transformation
  - Duplicate removal
  - Standardisation
  - Archiving
  - Export data marts

Populate

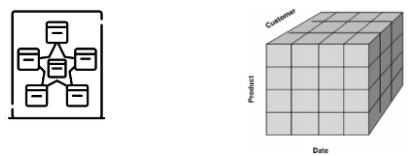
Replicate

Retrieve

Data warehouse presentation server

- Data mart
  - Subject-oriented
  - Implemented locally
  - Dedicated to a group of users
  - Refreshed regularly
  - Shared dimensions
  - Dimensional model
  - OLAP query services

ROLAP and/or MOLAP



Feed

Feedback portal

Ad hoc query tools

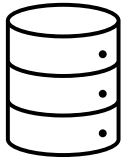
Report generators

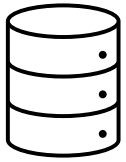
User applications

Modelling application
- Forecasting
- Data mining
- Budgets
- …

BI elements
Source systems

Source systems

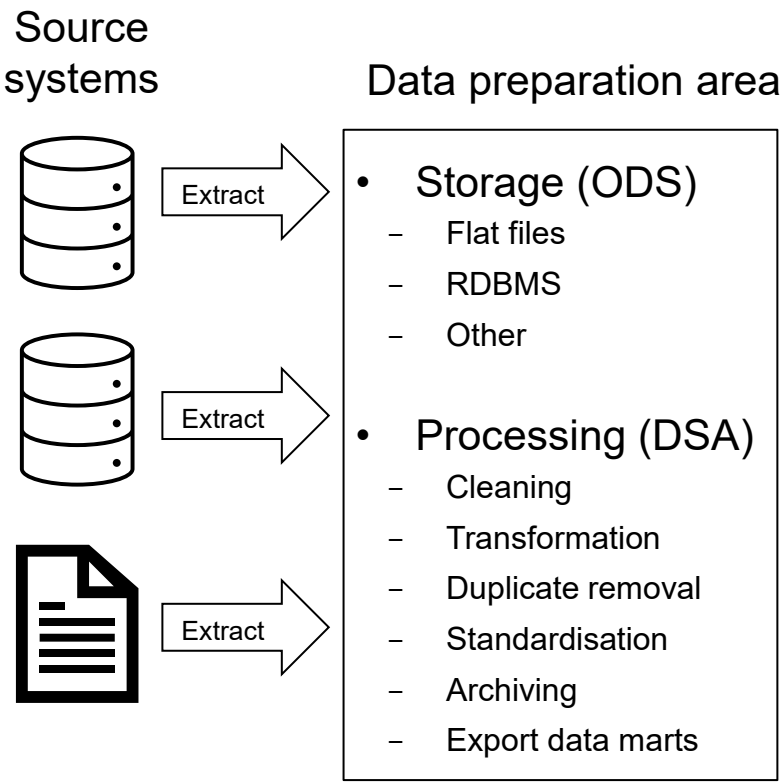






- "Operational recording system, whose function is to capture transactions related to the activity"
- Priority = uptime
- Few queries
- Little history
- Little effort required to standardise basic dimensions
- Operational keys that ensure the uniqueness of certain elements

BI components
Data preparation area

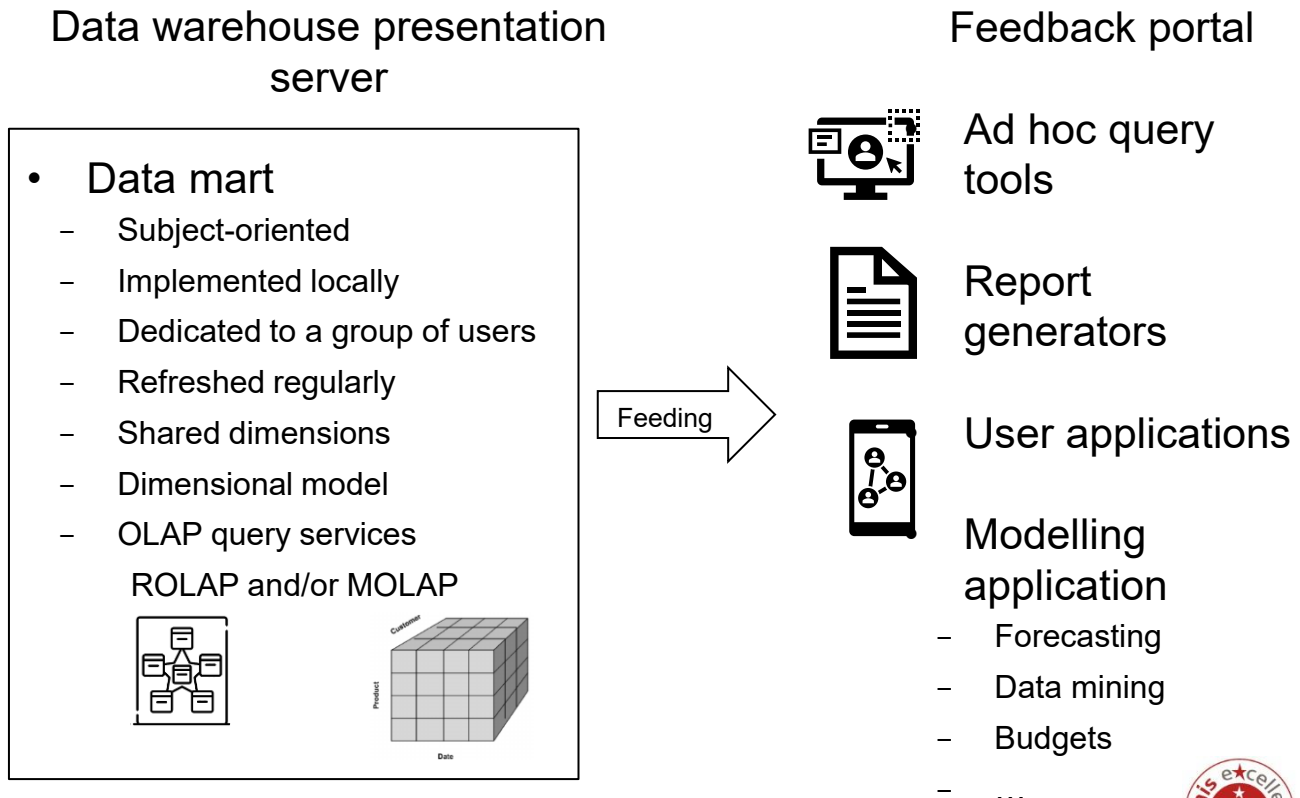


- "All processes that clean, transform, combine, archive, and remove duplicates, i.e., prepare source data for integration and subsequent use within the data warehouse."
- No user access
- ODS = Operational Data Store
  - Identical storage of operational data
  - Consolidation of different sources
  - Common basis for calculations
- DSA = Data Staging Area
  - Working area
  - Addition of calculated fields
  - Temporary tables

BI elements

Data warehouse presentation server

- "Target machine on which the DWH is stored and organised to respond directly to queries issued by users, report generators and other applications"
- Data mart = subset of a DWH according to a business process. Feasible project
- Dimensional model
  - Dimensions
  - Fact table
- ROLAP = Relational database
- MOLAP = Cube: Proprietary database



BI elements
Reporting portal

- Query tools: invites users to formulate their own queries by directly manipulating relational tables and their joins
- Report generators: predefined reports distributed automatically (scheduling/broadcasting) or viewed on demand
- User application: "Set of tools that query, analyse and present information to meet a specific need". Consists of data access tools, spreadsheets and graphics software
- Modelling application

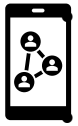
Feedback portal



Ad hoc query tools



Report generators



User applications

Modelling application

- Forecasting
- Data mining
- Budgets
- …